

Vortex-induced vibration dynamic characteristics monitoring with AI-based target object detection for long-span bridges

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ABSTRACT

This research proposes an innovative framework for monitoring bridge Vortex-induced Vibration (VIV) dynamic characteristics utilizing AI-based machine-learning target object detection and tracking techniques with keypoint detection. Two camera calibration methods are implemented for cases with and without intrinsic and extrinsic camera setup information based on homography matrix conversion and distance-based conversion. The framework is verified on a video recording of a real-bridge VIV event and a simulation animation of bridge VIV with a peak/trough-based statistical method for calculating VIV frequency and amplitude. Accurate detection is achieved in both cases with short video durations. The framework demonstrates great potential for real-time bridge VIV monitoring, requiring minimal camera calibration and a straightforward device setup. It offers reliable and accurate results while remaining cost-effective.

1. Introduction

Vortex-induced Vibration (VIV) is a phenomenon that occurs when a slender structure, such as a long-span bridge, experiences oscillations due to the bridge girder's resonance with the shedding of vortices in the wake of the girder. The VIV is a self-limiting vibration and is usually not considered a catastrophic threat to the integrity of the structures. However, the VIV happens under low wind speeds and lasts relatively long. Therefore, it can potentially cause structural fatigue and interrupt structural serviceability. Successful real-time monitoring of VIV could aid efficient and timely vibration control. Since the VIVs are usually single-mode vibrations, when the dynamic characteristics of the specific modes of the VIVs are recognized in their early stages, it is possible to perform active vibration control by activating selected pre-installed local vibration control techniques such as piezoelectric actuators [1], PID-controlled actuators [2,3], dampers [4], etc. to prevent the VIV from developing.

Currently, the commonly adopted method for monitoring bridge VIV is by installing acceleration and displacement sensors at a limited number of locations on bridges, which can be costly and require constant maintenance and calibration. In addition, the installations and calibrations of the sensors are time-consuming and lack the quick-response

capability, resulting frequently in inadequate monitoring of the entire bridge VIV occurrence. As a result, sustained, comprehensive, and cost-effective structural monitoring for bridge VIV remains an open issue.

In recent years, the implementation of computer vision methods in the field of structural health monitoring (SHM) has reshaped the landscape of structural assessment and maintenance [5,6]. In the field of bridge monitoring, scholars have explored the application of computer vision as well [7]. Pan et al. [8] developed a video deflectometer using the off-axis Digital Image Correlation (DIC) technique for real-time, non-contact, and targetless measurement of vertical deflection of bridges. Dong et al. [9] presented a vision-based structural dynamic displacement measurement method based on multi-point pattern matching and verified the result with onsite measured data from accelerometers. Li et al. [10] carried out onsite bridge VIV detection based on video inputs with DIC techniques and obtained sub-pixel level deformation information of the bridge girder. Zhao et al. [11] performed bridge VIV monitoring with the Channel and Spatial Reliability Tracker (CSRT) technique at five different locations on the bridge girder and estimated the bridge VIV dynamic characteristics based on a hand-held camera recorded video. Lin et al. [12] proposed a vision-based modal analysis framework based on frequency domain decomposition capable of directly performing modal decomposition on vision data of bridges

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without needing to recover the velocity and displacement. Wang et al. [13] established a contactless vision-based modal testing framework based on gradient-based matching via voting (GMV) technique for identifying modal parameters of bridges in uncontrolled field conditions. The aforementioned methods provide sufficiently reliable results. However, these methods either require that the target objects remain in consistent environments, rendering them vulnerable to external environmental interference such as lighting/weather change and temporary blocking, or rely on multi-device configurations, increasing the susceptibility to systematic errors across devices. Therefore, these methods are less than ideal for sustained, cost-effective real-time bridge VIV monitoring over extended durations.

Based on the above insufficiencies, Shao et al. [14] proposed a monocular vision-based three-dimensional displacement measurement approach for structures that utilizes machine learning algorithms and conducted experiments on cantilever beams, which yielded good results. The method is promising in the field of structural health monitoring. Lu et al. [15] proposed a novel displacement measurement method based on machine-learning and digital image processing to mitigate the effects of ambient light changes. The method automatically extracts the calibration object in complex scenarios and precisely locates the calibration object under 24-hour ambient light changes. However, the above experiments are conducted in close camera-to-object distances under laboratory conditions and require accurate camera calibration, thus is unproven for onsite structural monitoring from a distance and for bridge monitoring videos without prior camera calibrations.

To summarize, existing CV-based methods eliminate the need for installation and calibration of physical sensors and provide satisfactory post-event monitoring results for VIV. However, they are prone to external interferences such as lighting/weather change, temporary blocking, and sudden blurriness, which are common in onsite real-time monitoring of bridges. Some machine-learning-based methods were proposed to adapt to the external interferences. They have been proven reliable and effective in controlled environments such as close-range indoor monitoring but are yet to be applied to real-time onsite bridge monitoring.

To address the abovementioned technical challenges, this research introduces an innovative framework for bridge VIV dynamic characteristic monitoring, driven by computer-vision and machine-learning-based target object detection and tracking techniques with keypoint detection. Due to its machine-learning nature, this framework exhibits robustness and flexibility in the presence of extrinsic interferences, such as temporary blocking, sudden blurriness, and bridge environmental variations in weather and lighting conditions affecting target objects. The framework only requires a single camera and an onsite or remote processing device setup, thus making it suitable for sustained and cost-effective real-time bridge monitoring. It also offers versatile applicability and is adaptive to scenarios with or without camera setup information through the utilization of camera calibration with homography matrix conversion and distance-based conversion methods. A peak/trough-based statistical method for calculating VIV frequency and amplitude is proposed to improve the accuracy of VIV dynamic analysis recognition for videos with short durations. Through validation, the method's effectiveness in both real-bridge video recordings and simulated bridge VIV animations is verified using a single-camera setup and a Python-supported data processing device, affirming its usability and robustness, as well as the potential for real-time bridge VIV monitoring.

2. Detection and tracking based on machine learning

This framework utilizes open-source algorithms YOLOv8 [16] and ByteTrack [17] to perform target-object-based keypoint detection and subsequent keypoint tracking to achieve VIV dynamic characteristics monitoring in a video of a real bridge VIV phenomenon captured by a hand-held camera.

The target object detection and tracking tasks are performed in a

video of a VIV event on Humen Bridge in Guangdong, China, on May 5th, 2020 [18]. The Humen Bridge is a suspension bridge in Guangdong, China. It spans 888 m and was opened to traffic in 1997. For the detection task, the target objects used are the drainage outlets of the box girder on the side of the girder section, as shown in Fig. 1 and Fig. 2.

According to the design of the Humen bridge, the spacing between the drainage outlets along the girder is 12 m, and there are 73 outlets on each side of the main span girder. In the video where the detection and tracking task is performed, 48 outlets are visible throughout the 33.8-second video length, the movements of the drainage outlets in the video are sufficient to reveal the VIV dynamic characteristics of the bridge.

2.1. Object detection with detection boxes

2.1.1. Object detection box based on YOLOv8

The detection task is performed with the open-source algorithm YOLOv8, a computer vision model developed by Ultralytics that supports object detection tasks written on the Python framework.

YOLOv8 first uses a group of Convolutional-Neural-Network-based (CNN) machine learning models to develop a spatial feature of the target object from user-specified training and validation sets of labeled data. The model would then plot different sizes of grids to detect all objects in a picture or a frame in a video and utilizes the Non-Maximum Suppression (NMS) technique to filter out untargeted detection boxes by assessing their Intersection-Over-Union (IOU) value with the spatial feature of the target object generated from the machine learning model, thus accomplishing the object detection task. The IOU value can be calculated with Eq.1.

$$\text{Intersection over Union (IOU)} = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

where A and B represent the area of the two detection boxes being assessed.

The YOLOv8 scans the picture or frame only once, hence the name YOLOv8 (You Only Look Once, 8th version). Fig. 3 shows the flow chart describing the training and detection process above.

2.1.2. Drainage outlet detection

This framework first creates detection boxes for clearly visible drainage outlets in the video to achieve computer-vision-based VIV dynamic characteristic monitoring.

For the training of the machine learning models, twenty snapshots of the video are taken, and the drainage outlets in each of the snapshots are manually labeled with rectangular boxes using the labeling software LabelMe [19], which records the center point coordinates, width, and height of each labeled box. The information from the twenty snapshots and their labeled boxes are randomly assigned to training sets (twenty snapshots) and validation sets (eight snapshots). The CNN-based

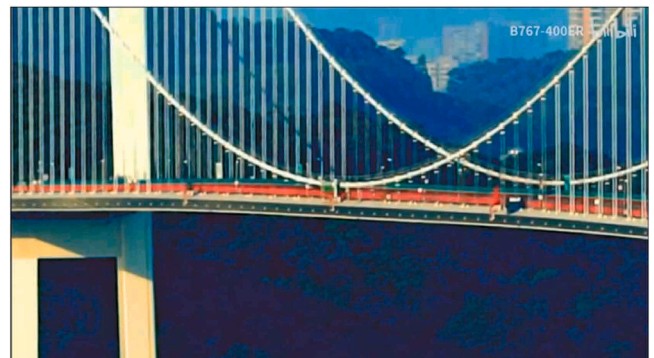


Fig. 1. Drainage outlets on the side of the girder section.

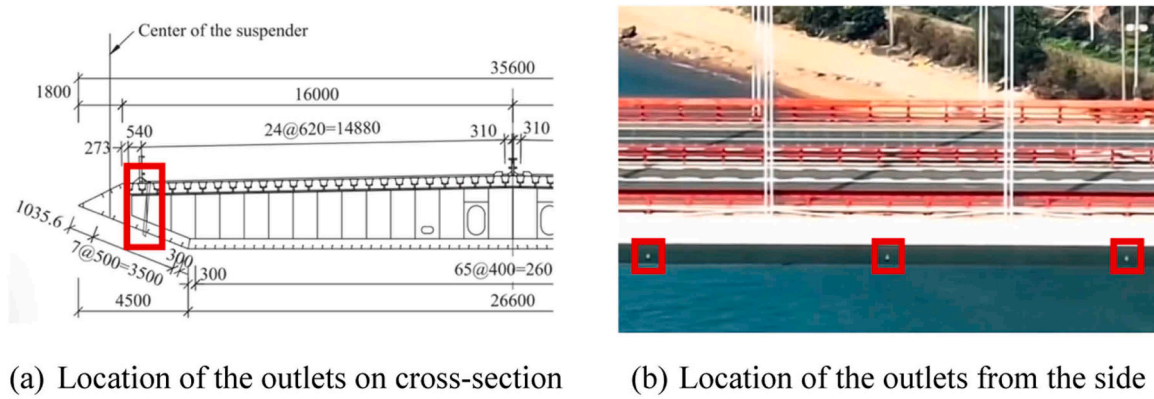


Fig. 2. Drainage outlets position on the cross-section of the girder.

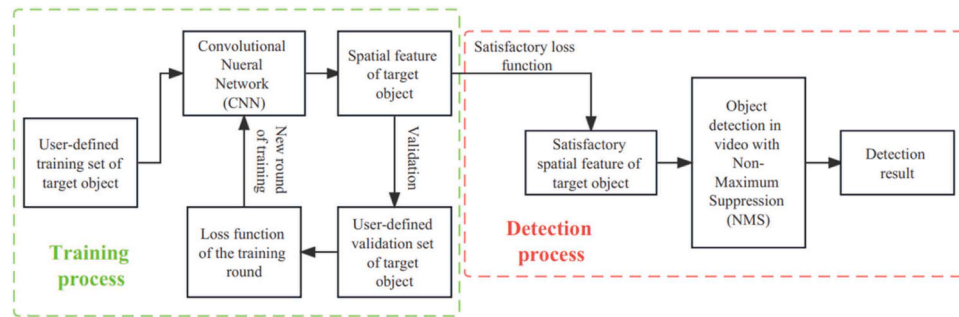


Fig. 3. Flow chart for object detection based on YOLOv8.

machine learning model would then develop an initial spatial feature of the drainage outlets based on the labeled outlets in the training sets and compare it with the labeled outlets in the validation sets to generate a loss function that reflects how well the initial spatial feature describes the appearance of the drainage outlets in the validation set. The model would then fine-tune the parameters within the convolutional neural network until the loss function trends toward a positive direction, indicating that the spatial feature description of the drainage outlets is becoming more accurate, finishing a round of training. Fig. 4 shows an example of the labeling of drainage outlets with rectangular boxes.

For training of the drainage outlets, 1000 rounds of training are performed, which produces a spatial feature of the drainage outlets with a model performance that indicates a satisfactory matching probability of the model's perception of drainage outlets compared to the labeled outlets in the validation sets. The performance of the model is described by the precision and mAP (mean average precision) values in YOLOv8, which are calculated with Eq. 2 and Eq. 3.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

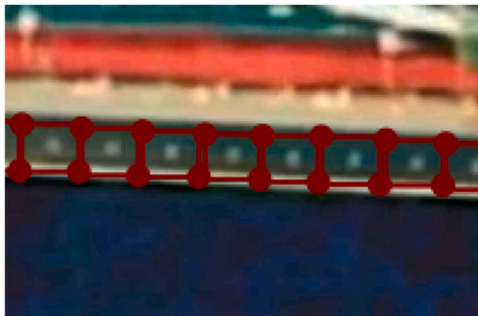


Fig. 4. Labeling of outlets with rectangular boxes.

where TP indicates True Positive and FP indicates False Positive. A true positive means the model correctly predicts an outlet, and a false positive means the model incorrectly predicts an outlet where it does not exist.

$$mAP = \frac{1}{n} \sum_{k=1}^n AP_k \quad (3)$$

where n indicates the number of classes, AP_k indicates the average precision (AP) of class k .

The model that satisfactorily describes the spatial feature of the drainage outlets from machine learning is then input into the detection model of YOLOv8 to identify detection boxes of drainage outlets in each frame of the video.

Utilizing detection box information, the spatial ranges of drainage outlets in each frame are estimated; however, the center point of each detection box may not precisely denote the location of each drainage outlet. Inaccuracy stems partly from manual labeling during the machine learning phase. Moreover, the detection model primarily identifies an object with a white center against a dark background. These factors contribute to inaccuracies in detection box center locations and inconsistent sizes of identical detection boxes across frames, resulting in errors in pinpointing the precise locations of drainage outlets across different frames.

To mitigate this inaccuracy, integrating keypoint detection within each detection box can be beneficial. By incorporating manually marked keypoints positioned at the center of each drainage outlet during the labeling stage using LabelMe within the machine learning process, the detection model could yield more precise outcomes that reflect the outlet coordinates in each frame. Fig. 5 illustrates the labeling and detection of outlets with keypoints within rectangular boxes, showcasing the potential enhancement in accuracy. Additionally, this approach could facilitate more reliable and consistent detection of

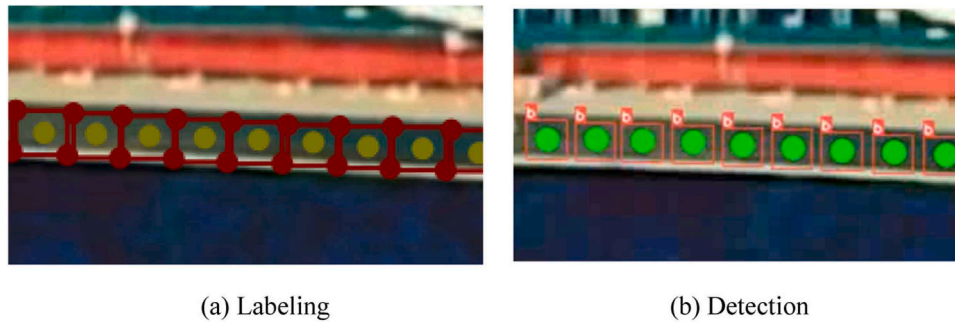


Fig. 5. Labeling and detection with keypoints and rectangular boxes.

drainage outlets across various frames, enhancing the overall performance of the detection model.

For the detection of drainage outlets with keypoint and detection box, the detection box would first provide an approximate range in which the outlet is located, then the keypoint would give more accurate coordinates of the center of the outlet in the frame that could yield more reliable results for VIV dynamic characteristics monitoring.

2.2. Tracking

With the drainage outlets detected using detection boxes and keypoints, the next step is to track the movements of each drainage outlet in the video. An open-source algorithm named ByteTrack is used for object tracking in this framework.

ByteTrack assigns a unique ID for all detection boxes in the first frame of the video, then utilizes the Kalman filter [20] and the Hungarian algorithm [21] to maintain the same unique box ID for each box throughout the duration of the video. The Kalman filter extracts the temporal feature of a detection box and predicts their coordinates in the following frame, then uses the Hungarian algorithm to calculate and compare the IOU value based on the predicted location of the detection box and the location of all detection boxes in the following frame to identify the same detection box, thus maintaining the same unique ID for the box from frame to frame. Each keypoint in the detection box inherits the same unique ID from the detection box. Fig. 6 shows a schematic for tracking the same detection box across two adjacent frames with ByteTrack and YOLOv8.

Bytetrack uses YOLOv8 as the detection platform and then compares the IOU value to assign unique IDs for the same detection boxes across different frames. The tracking of keypoints follows a similar pattern

within their respective detection boxes.

The tracking of detection boxes and keypoints is robust and adaptive to temporary blocking and graphical interference, such as lighting condition change, sudden blurriness, and change of color tone of the video, making it ideal for real-time monitoring, which is prone to environmental interference, as shown in Fig. 7.

As seen In Fig. 7(a), the detection boxes remain consistent when the lighting condition changes. In Fig. 7(b), when a sudden blurriness occurs for the video, the detection boxes would recover quickly after the resolution of the video recovers. In Fig. 7(c), the detection boxes remain consistent when the color tone changes from color to black-and-white. In Fig. 7(d), when the detection boxes are temporarily blocked, the detection boxes would recover quickly after the blocking.

3. VIV monitoring based on uncalibrated real bridge video

With the detection and tracking of drainage outlets introduced in the previous section, the time history of the two-dimensional coordinates of each drainage outlet throughout the duration of the video can be obtained. By extracting the coordinates of the detected keypoints in each frame, the VIV shape of the bridge can be visualized, as shown in Fig. 8.

In Fig. 8, the y-direction amplitudes of all detected outlets at a time point are shown in their fraction of the height of the video. With further post-processing conducted in MATLAB [22], the coordinates of drainage outlets can be used to reflect the VIV dynamic characteristics of the bridge, such as frequency, amplitude, and mode shape. On bridges where drainage outlets are not visible or installed, other structural/non-structural components such as roadside lamps, hangers, and crash barriers can be used as the target objects.

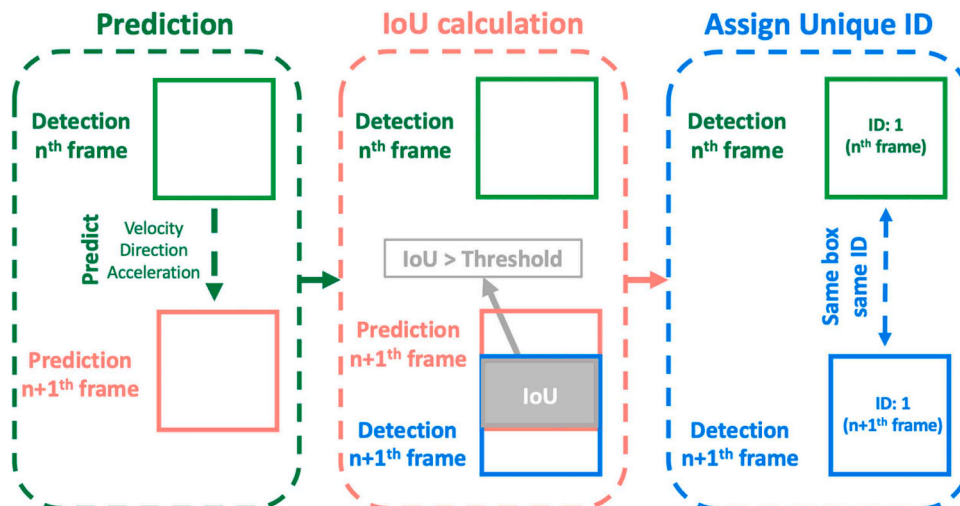


Fig. 6. Tracking of detection boxes and keypoints.

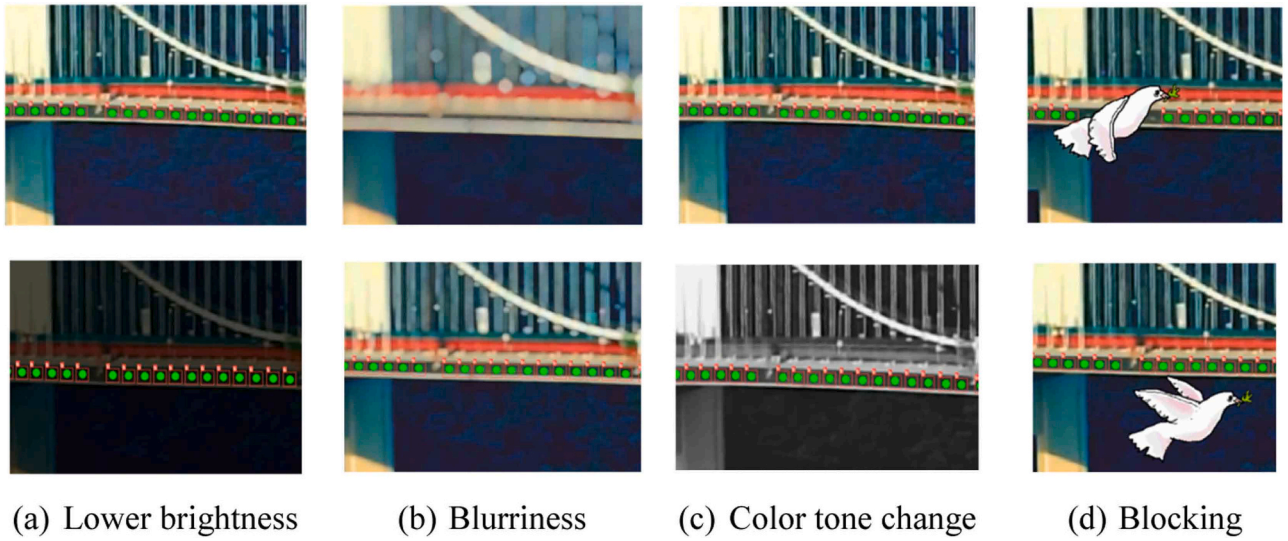
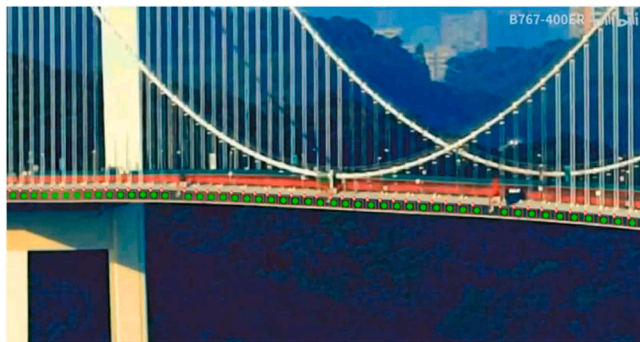
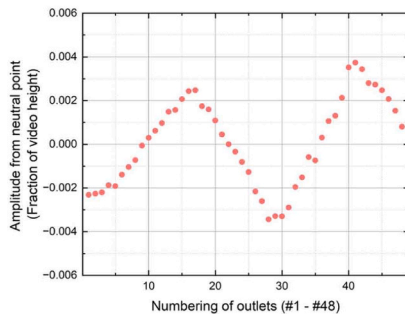


Fig. 7. Interference robustness of the tracking method.



(a) Keypoint tracking



(b) Extracted result in MATLAB

Fig. 8. Bridge VIV visualization.

3.1. Distance-based conversion

As seen in Fig. 8(a), the video is taken from an angle respective to the plane of the side of the girder. Based on the geometric perspective principle, the drainage outlets that oscillate with the same amplitude at different distances from the camera exhibit different object dimensions and amplitude because of the zooming effect origins from their respective distances to the camera. The ratio of this zooming effect follows a linear relationship with respect to the outlets' distances from the camera.

To adapt the zooming effect for videos taken from an angle, the zooming ratios of different outlets in the video need to be calculated. To calculate the zooming ratio of the drainage outlets located from the left

to right side of the video, the distances between adjacent keypoints on the x-direction are extracted where the distances are normalized respective to the x-direction distance between the two keypoint furthest to the right of the video. Eq. 4 indicates the calculation of the zooming ratio.

$$R_n = \frac{x_{n+1} - x_n}{x_{48} - x_{47}} \quad (4)$$

where R_n indicates the zooming ratio around the n^{th} outlet in a frame, x_n is the median x-coordinate of the n^{th} outlet, x_{47} and x_{48} are the median x-coordinate of the two rightmost outlets.

The zooming ratio can be used to normalize the oscillation amplitude of all drainage outlets to a unified scale in the video. A schematic comparison between the ununified coordinates directly from the keypoints detection and the coordinates from the unified scale keypoint detection in the video coordinates system is shown in Fig. 9.

In Fig. 9, the keypoint coordinates of the outlets are depicted before and after normalization, with red dots representing ununified coordinates and blue dots representing unified coordinates adjusted with the zooming ratio. This normalization process mitigates the impact of geometric perspective, ensuring that all outlet amplitudes are standardized to the same scale. As illustrated in Fig. 9, outlets farther from

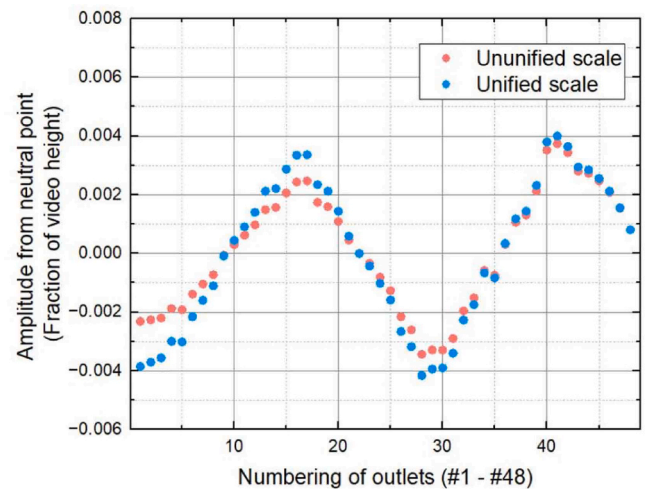


Fig. 9. Keypoints coordinates from unified and ununified keypoint detection.

the camera (on the left side of the figure) exhibit greater amplification compared to those closer to the camera (on the right side of the figure). Following normalization, the amplitudes of all outlets become more consistent and provide a realistic basis for amplitude comparison on a unified scale.

From the unified scale, the oscillation amplitude in the world coordinate system can be calculated with Eq. 5.

$$y_{n,w} = \frac{y_{n,v}}{R_n} \times \frac{d}{x_{48} - x_{47}} \quad (5)$$

where $y_{n,w}$ is the y-direction amplitude of the n^{th} outlet in the world coordinate system, $y_{n,v}$ is the y-direction amplitude of the n^{th} outlet in the video coordinate system, d is the spacing between drainage outlets in the world coordinate system, which equals to 12 m in the video analyzed.

The dynamic characteristics of the VIV can be calculated with the keypoint coordinates from the world coordinate system.

3.2. Frequency determination

With the time histories of the y-direction movements of all keypoints detected, it is possible to calculate the VIV frequency of the girder section. The most straightforward method for such calculation is converting the time histories of y-direction movements into the frequency domain with the Fast Fourier Transformation (FFT) method. The predominant frequency of the keypoints movements is the VIV frequency of the girder. However, since the video analyzed has a short duration, the sample size for FFT is limited, and the direct result is that the frequency distribution (x-axis of the FFT) is relatively coarse. With the 33.8-second video duration and 60 fps frame rate, the FFT calculation yields a frequency step size of 0.0296 Hz. Conducting FFT calculations for movements of all 48 detected keypoints, the predominant frequencies are 0.3550 Hz for all keypoints detected, indicating that the frequencies for each keypoint all fall within the 0.0296 Hz range around the 0.3550 Hz frequency. Therefore, the FFT calculation is unsuitable for calculating accurate VIV frequency results based on videos with short durations.

To solve such inaccuracy in the VIV frequency calculation, this framework proposes a statistical method with peak-to-peak and trough-to-trough time duration measurement. Fig. 10 shows how the peak-to-peak and trough-to-trough time durations are measured. Eq. 6 shows the calculation method.

$$\begin{cases} g_{p,n} = t_{p,n+1} - t_{p,n} \\ g_{t,n} = t_{t,n+1} - t_{t,n} \end{cases} \quad (6)$$

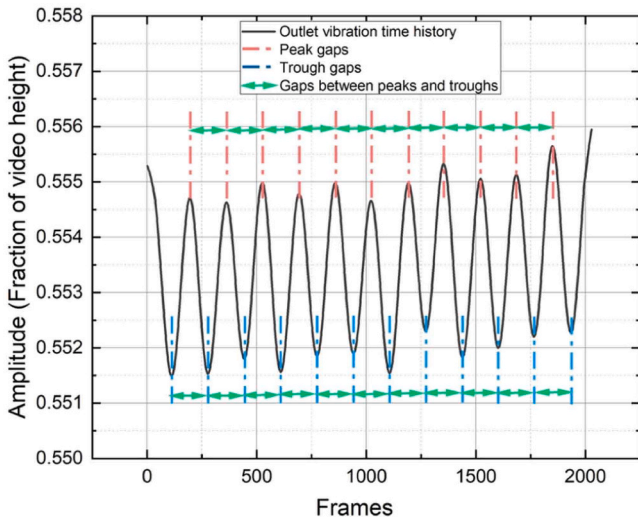


Fig. 10. Schematic for peak/trough time durations.

where $g_{p,n}$ and $g_{t,n}$ are the gap between n^{th} and $n + 1^{th}$ peaks and between n^{th} and $n + 1^{th}$ troughs, $t_{p,n}$ and $t_{t,n}$ are the time point where the n^{th} peak or trough occurs in the y-direction oscillation time history for a drainage outlet.

By gathering statistics on the time durations between adjacent peaks/troughs from the movement time history of all detected outlets, it is observed that the distribution of the time durations follows a normal distribution, as shown in Fig. 11.

In Fig. 11, 989 samples are included. The adjacent peak/trough time durations distribution is a relatively tight normal distribution with a small kurtosis value and slight negative skewness, indicating light tails and approximate symmetry on the normal distribution. The mean value of the peak/trough time duration is 165.175 frames, the median value is 165 frames, and the standard deviation is 7.023 frames. The 95% confidence interval is [164.737, 165.613]. The frequency of the VIV can be calculated with the frame values using Eq. 7.

$$f = \frac{fps}{n_{frame}} \quad (7)$$

where f is the frequency in Hz, fps is the frame-per-second value of the video, which is 60 fps for the video analyzed, n_{frame} is the time duration frame value.

The calculated frequency information based on the adjacent peak and trough time duration frame values and the comparable frequencies determined by Li et al. [10] and Zhao et al. [11] for the same VIV event is summarized in Table 1.

Based on the mean value of the distribution of adjacent peaks and troughs, this research determines the VIV frequency to be 0.3633 Hz, which slightly differs from the existing literature. This discrepancy in results could be attributed to the differing frequency calculation approaches as well as the different numbers of target objects used.

In their study, Zhao et al. [11] and Li et al. [10] analyzed the movement of a limited number of target objects at various positions along a span in a 30-second video. Zhao et al. employed a Bayesian Fast Fourier Transform (FFT) method to calculate the oscillation frequencies of five target objects, and Li et al. utilized FFT for a single target object. Given the brevity of the video duration, FFT analysis may yield a coarse frequency axis in the frequency domain.

To mitigate this limitation, this research adopts an alternative approach by summarizing the time differentials between adjacent peak and trough occurrences for all 48 detected target objects. Distribution analysis reveals a clear and symmetric normal distribution of these time differences, which provides a more accurate representation of the true

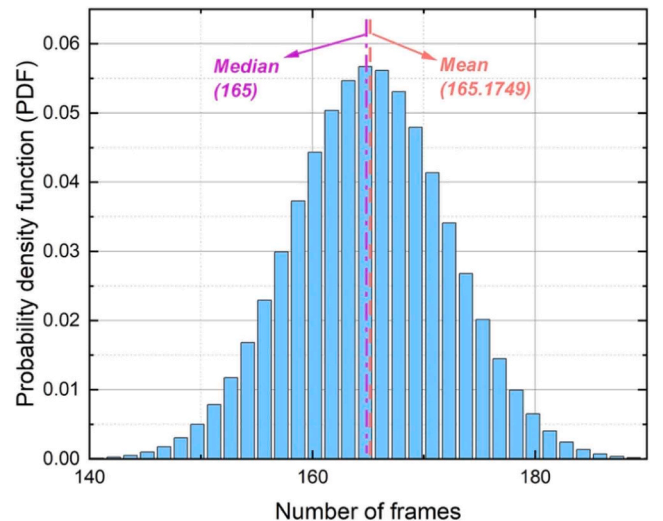


Fig. 11. Distribution of adjacent peak and trough time duration.

Table 1

VIV frequency based on adjacent peak and trough time durations.

	Mean	Median	95 % CI	Std. Dev.
Frequency (Hz)	0.3633	0.3636	[0.3623, 0.3642]	0.0154
Li et al. [10] (Hz)	0.3650	-	-	-
Zhao et al. [11] (Hz)	0.3687	-	-	-

oscillation frequencies of the target objects.

3.3. Amplitude determination

The time history of the y-direction movement of a detected keypoint in the video coordinates system is shown in Fig. 12.

As seen in Fig. 12, there is a clear upward trend of the oscillation movement of the keypoint over time. The trend is potentially due to the video's camera movement originating from the recording's hand-held nature. To minimize the effect of the camera movement on the calculation for VIV amplitude, a statistical method with peak-to-tough distance is proposed. Fig. 13 shows a schematic for such a method.

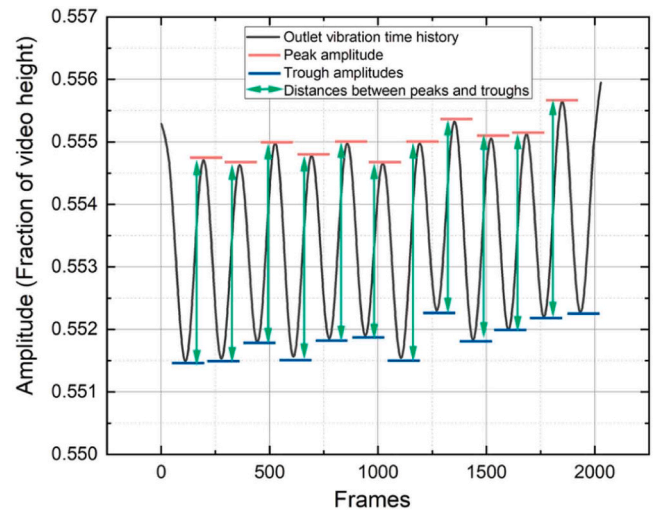
By measuring the y-direction distance between the adjacent peaks and troughs in the video coordinates system and gathering the statistics, the amplitude distribution for all detected keypoints can be calculated. Using Eq. 5, the amplitude distribution of the keypoints in the world coordinate system can be calculated. Fig. 14 shows the maximum amplitude of keypoints #10 to #48 in the world coordinate system.

In Fig. 14, the amplitude distribution of each keypoint detected over time is presented in a box plot format. The figure shows that the maximum amplitude occurs to the right of the mid-span. Near the 3/8 span and 5/8 span, there are two clear peaks as well. The maximum amplitude result of the VIV is summarized in Table 2.

As summarized in Table 2, the maximum amplitude occurred at keypoint #29, 18 m to the right of the mid-span of the bridge, and has a value of 60.21 cm. The y-direction resolution of the video (1080 pixels) controls the precision of the amplitude, and the value is determined to be ± 3.54 cm.

3.4. Mode recognition

In the study by Zhao et al. [11], the mode shape was determined manually by observing that the sections at 1/4 and 3/4 span exhibited the same phase, as did those at 5/8 and 3/8 span, with maximum vibration amplitude occurring at the 1/2 span position. In our study, a similar methodology was employed to identify the mode shape with more target objects, and the mode is determined as the 3rd order vertical

**Fig. 13.** Peak-to-trough method.

bending mode, which matches the mode shape result by Zhao et al. The mode shape from numerical analysis by Zhao et al. is shown in Fig. 15.

Furthermore, while Zhao et al. monitored only five target objects, our research extended the scope to include 48 target objects spanning the same range. Interestingly, it was observed that the mode shape did not exhibit symmetry with respect to the 1/2 span position. Instead, symmetry was noted to occur approximately 18 m to the right of the 1/2 span. This deviation from symmetry could potentially be attributed to the natural wear and tear experienced by the bridge during its 20 years of service.

4. 3D reconstruction based on a single camera and homographic conversion

4.1. Single-camera 3D reconstruction

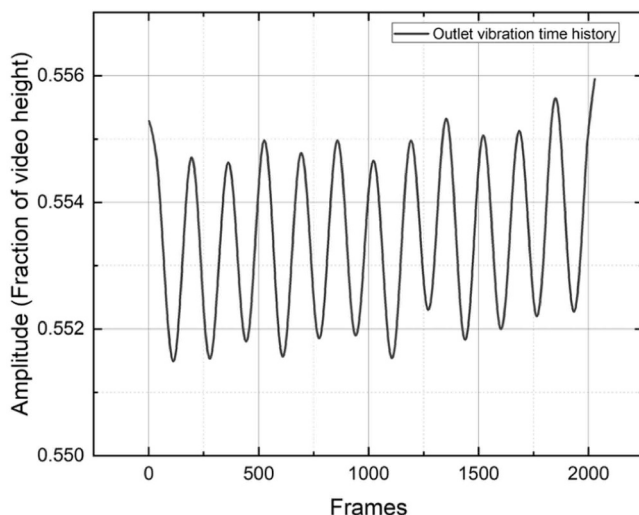
The distance-based conversion method introduced in the previous section provided sufficient accuracy when camera setup information is unavailable. To further improve the conversion accuracy between the video coordinates system and the world coordinate system when camera setup information is available, a three-dimensional reconstruction (3D reconstruction) camera calibration technique with a single camera setup is implemented.

The 3D reconstruction camera calibration technique utilized in this framework is single-camera calibration with intrinsic and extrinsic matrices, which associates the coordinates of pixels in a video with the coordinates of their counterparts in the real world. Three coordinate systems are used in this technique, including the video coordinate system (VCS), the camera coordinate system (CCS), and the world coordinate system (WCS). A schematic of the single camera calibration is shown in Fig. 16.

As seen in Fig. 16, the WCS is at an angle and a distance compared to the CCS. To convert the object's coordinates from the WCS to the CCS, an extrinsic matrix including nine rotation terms and three translation terms is implemented, as shown in Eq. 8.

$$\begin{bmatrix} x_c \\ y_c \\ 1 \end{bmatrix} = \begin{bmatrix} r_1 & r_2 & r_3 & t_1 \\ r_4 & r_5 & r_6 & t_2 \\ r_7 & r_8 & r_9 & t_3 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (8)$$

where x_c and y_c are the point's coordinates in the CCS, X_w , Y_w , and Z_w are the point's coordinates in the WCS, the r terms are the rotation components in the extrinsic matrix, and the t terms are the translation components in the extrinsic matrix.

**Fig. 12.** Time history of y-direction movement of keypoint #20.

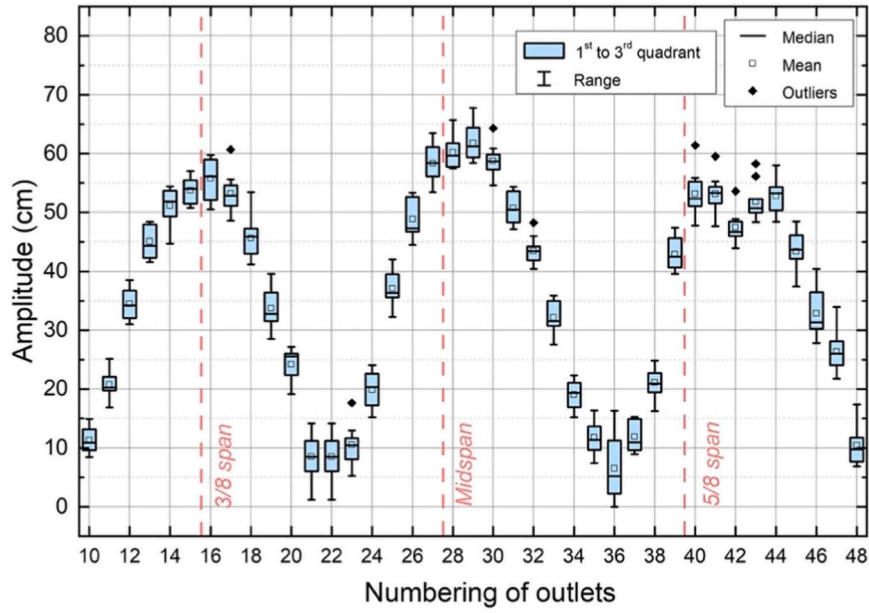


Fig. 14. Maximum amplitude distribution for all keypoints.

Table 2
maximum amplitude for keypoints detected.

Detection target	Location	Value (cm)
Amplitude (cm)	6 m to the right of 3/8 span	55.77
	18 m to the right of Mid-span	60.21
	6 m to the right of 5/8 span	53.12

By multiplying the coordinates of the object in the WCS by the extrinsic matrix, they are converted to the CCS. An intrinsic matrix is used to convert the object's coordinates from the CCS to the VCS. Eq. 9 shows the intrinsic matrix for a camera in which the optical axis and sensor axis are aligned.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_c/Z_c \\ y_c/Z_c \\ 1 \end{bmatrix} \quad (9)$$

where u and v are the point's coordinates in the VCS, Z_c is the z-direction distance between the point in the WCS and the CCS, f_x and f_y are the focal length of the camera on the x and y-direction, c_x and c_y are the coordinates of the optical center in the VCS.

Combining the extrinsic matrix and the intrinsic matrix, the conversion of coordinates of a point from the WCS to the VCS is shown in Eq. 10.

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_1 & r_2 & r_3 & t_1 \\ r_4 & r_5 & r_6 & t_2 \\ r_7 & r_8 & r_9 & t_3 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (10)$$

After the multiplication of extrinsic and intrinsic matrices, there are a total of eleven unknown parameters. Therefore, solving problems requires at least six pairs of known coordinates from the WCS and the VCS. However, in the cases of drainage outlet movement detection in this

research, all drainage outlets are located in the same plane in the WCS. Therefore, the conversion can be simplified by setting the value of Z to zero in Eq. 10. The result of this simplification is the homography matrix conversion, as shown in Eq. 11.

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_1 & r_2 & t_1 \\ r_3 & r_4 & t_2 \\ r_5 & r_6 & t_3 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} \quad (11)$$

After multiplying extrinsic and intrinsic matrices in the homography matrix conversion, there are seven unknown parameters, and it requires at least four pairs of known coordinates from the WCS and the VCS to solve. The conversion accuracy by the homography matrix can be further improved by utilizing the least square method with more than four pairs of known coordinates.

In the previous section, the distance-based conversion provides sufficient conversion functionality between the WCS and the VCS for the purpose of this research. While the homography matrix conversion

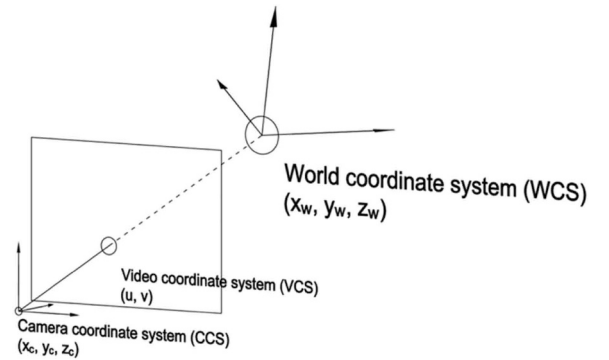


Fig. 16. Single camera calibration.

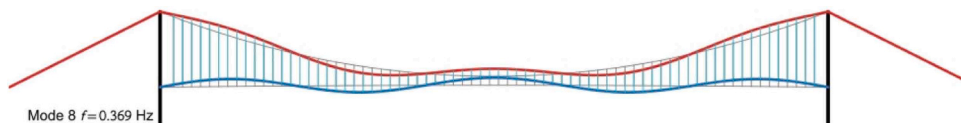


Fig. 15. 3rd order vertical bending mode shape by Zhao et al. [11].

requires additional information on the camera setup, including focal length and width/height of the video as well as at least four pairs of known coordinates from the world and video, it provides a more accurate, reliable, and systematic way for camera calibration, which is beneficial for accurate real-time monitoring application of bridge VIV dynamic characteristics monitoring.

4.2. Bridge VIV animation detection verification

To demonstrate the application of the abovementioned camera calibration methods on bridge VIV monitoring and verify their respective accuracies, an animation for a bridge under the influence of VIV is created in Blender [23]. The bridge in the animation is a suspension bridge with a span of 1000 m, vibrating with a 100 m wavelength at a frequency of 0.4774 Hz and with an amplitude of 2 m. Circular shapes that are similar to the shape of the drainage outlets are attached to the side of the girder at a spacing of 20 m to represent the outlets and to serve as the target objects being detected. A chessboard with nine pairs of known coordinates is added to the same z-direction plane of the target objects to serve as the basis for camera calibration with homography matrix conversion. The focal length of the virtual camera that captures the bridge VIV movements and the chessboard is 50 mm in both the x and y directions and is used for the intrinsic part calculation of the homography matrix. The 20 m distance between the outlets is used to calculate the zooming ratios for the distance-based conversion. The animation setup is shown in Fig. 17, and a snapshot of the captured video is shown in Fig. 18.

The video is recorded, and the drainage outlets on the side of the girder (target objects), as well as the chessboard, are put through the detection and tracking process introduced in the previous sections. For realistic purposes, blurriness is added to the video to simulate real-bridge monitoring conditions. Fig. 19 shows a schematic for the labeling of the drainage outlets and the chessboard with rectangular boxes and keypoints.

A schematic of the tracking result of the outlets in the animation is shown in Fig. 20.

The frequency and the amplitude of the VIV are calculated using the methods introduced in previous sections. Homography matrix conversion and distance-based conversion are used in the camera calibration process of the calculation. The result is shown in Table 3.

As seen in Table 3, the AI-based detection and tracking method results with the homography matrix conversion and the distance-based conversion match the actual VIV dynamic characteristics well. The calculation based on homography matrix conversion provided more accurate results than the distance-based conversion method. However, both methods provided reliable results with small errors for the VIV frequency calculation (0.13 % for homography matrix conversion, 0.25% for distance-based conversion) and the VIV amplitude calculation (0.14 % for homography matrix conversion, 0.61% for distance-based conversion). Based on the calculation results, it is concluded that when the camera setup is known, and prior camera calibration is possible, the homography matrix conversion provides better accuracy and reliability. In situations where these conditions are not met, such as



Fig. 17. Animation setup.

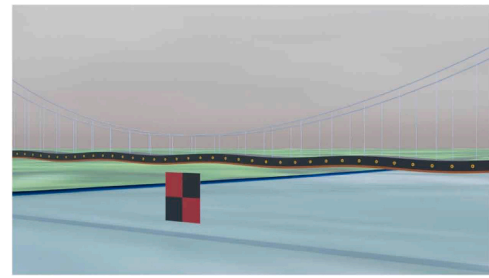


Fig. 18. A snapshot of the captured video.

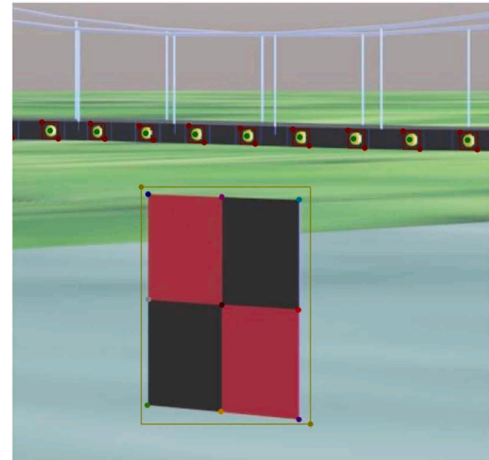


Fig. 19. Labeling of outlets and chessboard.

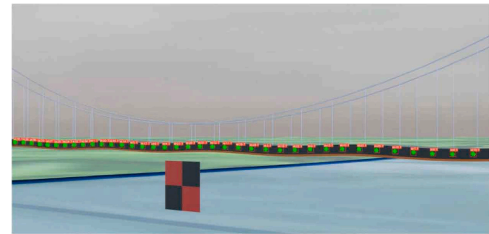


Fig. 20. Tracking of the outlets.

Table 3

Result comparison.

	Animation value	Homography matrix/ 95 % CI	Distance-based/ 95 % CI
Frequency (Hz)	0.4774	0.4768/ ± 0.0009	0.4762/ ± 0.0011
Amplitude (m)	2.0000	2.0027/ ± 0.0266	1.9878/ ± 0.0221

for dynamic characteristics monitoring for a bridge VIV event with an arbitrarily recorded video or a fast-response onsite monitoring of a bridge VIV event, the distance-based conversion could provide reliable results as well.

5. Conclusion

This research proposes an innovative framework for monitoring bridge VIV dynamic characteristics utilizing machine-learning-based target object detection and tracking techniques with keypoint detection. This framework exhibits robustness and flexibility in the presence

of external interferences and is capable of performing camera calibration with and without intrinsic and extrinsic camera setup information. A statistical method based on peak and trough analysis is used for calculating VIV frequency and amplitude, and its effectiveness is confirmed through verification on both real-bridge VIV event video footage and an animated representation of bridge VIV. Accurate detection is achieved in both scenarios, underscoring the efficacy of the proposed framework.

The highlight of this framework is summarized as follows:

1. The framework is adaptive to external environmental interference, such as temporary blocking, lighting/weather change, and short-term blurriness of the video.
2. A distance-based camera calibration estimation method is proposed in the framework, allowing accurate monitoring of the bridge VIV dynamic characteristics without requiring information regarding the camera's intrinsic and extrinsic setup.
3. A VIV frequency/amplitude determination method based on peak and trough analysis in the time history of target objects is developed. This technique enables precise determination of frequency and amplitude, even in short videos with camera motion.
4. The framework only requires a single-camera setup and a Python-supported processing device to achieve its functionality. This framework is suitable for simple, fast, reliable, and cost-effective real-time bridge VIV dynamic characteristics monitoring.

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CRedit authorship contribution statement

Mingjin Zhang: Conceptualization, Investigation, Writing – review & editing. **Jingxi Qin:** Conceptualization, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Fanying Jiang:** Investigation. **Renan Yuan:** Investigation, Writing – review & editing. **Zilong Ti:** Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Zhao W, Wang L. A novel interval dynamic topology optimization methodology of piezoelectric structures under reliable active control. *Comput Methods Appl Mech Eng* 2024;421(116766).
- [2] Liu J, Wang L. Hybrid reliability-based sequential optimization for PID vibratory controller design considering interval and fuzzy mixed uncertainty. *Appl Math Model* 2023;122:796–823.
- [3] Wang L, Zhao Y, Liu J. A Kriging-based decoupled non-probability reliability-based design optimization scheme for piezoelectric PID control systems. *Mech Syst Signal Process* 2023;203(110714).
- [4] Chen L, Liu Z, Nagarajaiah S, Sun L, Zhao L, Cui W. Vibration mitigation of long-span bridges with damped outriggers. *Eng Struct* 2022;271(114873).
- [5] Spencer B, Hoskere V, Narazaki Y. Advances in computer vision-based civil infrastructure inspection and monitoring. *Engineering* 2019;5(2):199–222.
- [6] Zhuang Y, Chen W, Jin T, Chen B, Zhang H, Zhang W. A review of computer vision-based structural deformation monitoring in field environments. *Sensors* 2022;22(10):3789.
- [7] Luo K, Kong X, Zhang J, Hu J, Li J. Computer vision-based bridge inspection and monitoring: a review. *Sensors* 2023;23(18):7863.
- [8] Pan B, Tian L, Song X. Real-time, non-contact and targetless measurement of vertical deflection of bridges using off-axis digital image correlation. *NDT E Int* 2016;79:73–80.
- [9] Dong C, Ye X, Jin T. Identification of structural dynamic characteristics based on machine vision technology. *Measurement* 2018;126:405–16.
- [10] D. Li, W. Chen, B. Peng, B. Yan, "Rapid DIC deformation measurement technology and engineering application," 25 6 2021. [Online]. Available: (<http://www.zbl.cn/index.php?m=content&c=index&a=show&catid=14&id=725>). [Accessed 26 9 2023].
- [11] Zhao L, Cui W, Shen X, Xu S, Ding Y, Ge Y. A fast on-site measure-analyze-suppress response to control vortex-induced-vibration of a long-span bridge. *Structures* 2022;35:192–201.
- [12] G. Lin, W. Li, Z. Lu, Output-only modal analysis of the humen bridge from video measurement, In: *Proceedings of the 14th International Conference on Damage Assessment of Structures*, Online, 2022.
- [13] Wang M, Ao W, Bownjohn J, Xu F. Completely non-contact modal testing of full-scale bridge in challenging conditions using vision sensing systems. *Eng Struct* 2022;282(114994).
- [14] Shao Y, Li L, Li J, Li Q, An S, Hao H. Monocular vision based 3D vibration displacement measurement for civil engineering structures. *Eng Struct* 2023;293(116661).
- [15] Lu B, Bai B, Zhao X. Vision-based structural displacement measurement under ambient-light changes via deep learning and digital image processing. *Measurement* 2023;208(112480).
- [16] G. Jocher, A. Chaurasia and J. Qiu, YOLO by Ultralytics (Version 8.0.0) [Computer software]. (<https://github.com/ultralytics/ultralytics>), 2023.
- [17] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, et al., ByteTrack: Multi-Object Tracking by Associating Every Detection Box, in *ECCV 2022*, Tel Aviv, 2022.
- [18] B767–400ER, "VIV of Humen Bridge on 2020/5/5," Bilibili, 5 5 2020. [Online]. Available: (https://www.bilibili.com/video/BV1Nt4y1178T/?spm_id_from=333.337.search-card.all.click&vd_source=96a4900b6cb39e77593852dc8ff59455). [Accessed 26 9 2023].
- [19] W. Kentaro, "Labelme: Image Polygonal Annotation with Python," [Online]. Available: (<https://github.com/wkentaro/labelme>).
- [20] Kalman R. A new approach to linear filtering and prediction problems. *ASME J Basic Eng* 1960;82:35–45.
- [21] Kuhn HW. The Hungarian Method for the assignment problem. *Nav Res Logist Q* 1955;2:83–97.
- [22] MATLAB, 9.7.0.1190202 (R2019b), Natick, Massachusetts: The MathWorks Inc., 2018.
- [23] Blender Foundation, Blender, Amsterdam: Blender.org, 2023.